SIMATS ENGINEERING
ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I s.sri mathi (192424095) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Sri Mathi

Annexure A

APPLICATION- BASED QUESTIONS

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

Demonstrate how the encryption process transforms the message step-by-step.

Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

Analyze how investigators could detect the presence of hidden data within the image.

Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

QUESTION 1: Secure Messaging System Using Keyword-Based Encryption

1.Scenario

A company uses a **keyword-based encryption technique** to protect internal communication. An employee wants to send the message **"HELLO"** using the keyword **"KEY"**.

The keyword-based technique used here is the **Vigenère cipher**, which is a classical **symmetric encryption technique**. The same keyword is used by the sender for encryption and by the receiver for decryption.

2.Objective

To demonstrate how the plaintext message **"HELLO"** is converted into ciphertext using the repeating keyword **"KEY"**.

3.Encryption Technique

The Vigenère cipher assigns numerical values to letters:

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

The encryption formula is:

$$C=(P+K) \bmod 26$$

Where:

- **C** = Ciphertext value
- **P** = Plaintext value
- **K** = Keyword value

Step 1 – Write the Plaintext

The employee wants to send:

Plaintext:

HELLO

Convert each letter into its numerical value:

Plaintext	**H**	**E**	**L**	**L**	**O**
Value	7	4	11	11	14

Step 2 – Repeat the Keyword

The keyword is:

KEY

The plaintext contains **5 characters**, but the keyword contains only **3 characters**.

Therefore, the keyword is repeated until it matches the length of the message:

KEYKE

So:

Plaintext	**H**	**E**	**L**	**L**	**O**
Keyword	K	E	Y	K	E

Numerical values:

	H	**E**	**L**	**L**	**O**
Plaintext	7	4	11	11	14

Step 3 – Perform Encryption

Using:

$$C=(P+K)\bmod 26$$

Character 1: H

$$(7+10)\bmod 26=17$$

17 = **R**

Character 2: E

$$(4+4)\bmod 26=8$$

8 = **I**

Character 3: L

(11+24)mod26=35mod26=9

9 = **J**

Character 4: L

(11+10)mod26=21

21 = **V**

Character 5: O

(14+4)mod26=18

18 = **S**

Therefore:

Ciphertext = RIJVS

4.Complete Encryption Table

Position	Plaintext	P Value	Keyword	K Value	Calculation	Ciphertext
1	**H**	**7**	**K**	**10**	**(7+10) mod 26 = 17**	**R**
2	**E**	**4**	**E**	**4**	**(4+4) mod 26 = 8**	**I**
3	**L**	**11**	**Y**	**24**	**(11+24) mod 26 = 9**	**J**
4	**L**	**11**	**K**	**10**	**(11+10) mod 26 = 21**	**V**
5	**O**	**14**	**E**	**4**	**(14+4) mod 26 = 18**	**S**

5.Final Result

Plaintext:

HELLO

Keyword:

KEY

Repeated Keyword:

KEYKE

Ciphertext:

RIJVS

6.How the Repeating Keyword Influences Each Character

The keyword does not remain fixed at only the first three characters. It is **repeated cyclically** to encrypt the entire message.

The pattern is:

KEY → KEY → KEY → ...

For the five-character message:

- HELLO
- KEYKE
- RIJVS

Therefore:

- **H** is shifted using **K**
- **E** is shifted using **E**
- **L** is shifted using **Y**
- The fourth character uses **K** again because the keyword has reached its end.
- The fifth character uses **E** again.

Thus, the keyword determines the shift applied to every plaintext character. Different keyword characters produce different shifts, making the ciphertext less directly related to the plaintext than a simple Caesar cipher.

7.Decryption

The receiver can recover the original message using the same keyword.

The decryption formula is:

$$P=(C-K)\mod 26$$

For example:

$$R-K=(17-10)\mod 26=7=H$$

Similarly:

- I − E = E
- J − Y = L
- V − K = L
- S − E = O

Therefore:

RIJVS → HELLO

8.Result

The plaintext **“HELLO”**, encrypted using the keyword **“KEY”**, produces the ciphertext:

RIJVS

The keyword is repeated as necessary, producing **KEYKE**, and each keyword character determines the amount by which the corresponding plaintext character is shifted.

QUESTION 2: Suspicious Digital Asset Investigation

1.Scenario

During a digital forensic investigation, investigators suspect that a company logo contains **confidential information hidden inside the image**.

The image looks normal when opened normally. There are no obvious visible changes, but confidential information may have been embedded using a technique called **steganography**.

Steganography is different from encryption:

- **Encryption** hides the meaning of information.
- **Steganography** hides the existence of the information itself.

Therefore, investigators need to determine whether the apparently normal image contains additional hidden data.

2.Objective

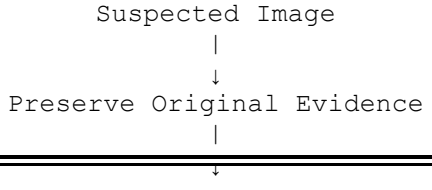
The objectives of the investigation are:

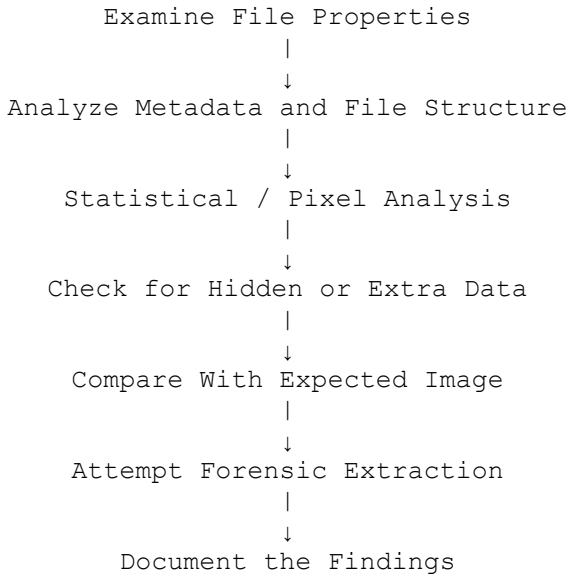
1. Determine whether hidden information exists inside the image.
2. Identify possible steganographic techniques used.
3. Analyze the image for abnormal patterns or additional data.
4. Determine whether the hidden information can be extracted.
5. Recommend techniques to make hidden communication more resistant to detection.

3.Investigation Process

A digital forensic investigator should follow a systematic process rather than immediately modifying or extracting the suspected data.

Basic Investigation Flow





Step 1 – Preserve the Original Image

The first step is to preserve the original evidence.

Investigators should:

- Make a forensic copy of the image.
- Avoid editing the original file.
- Record the file name and size.
- Calculate a cryptographic hash such as SHA-256.
- Perform analysis on a working copy.

This helps demonstrate that the evidence was not accidentally modified during the investigation.

Step 2 – Examine File Properties

Investigators can first examine basic properties such as:

- File type
- File size
- Image dimensions
- Color depth
- Compression information
- Creation/modification information
- Metadata

For example, if a small company logo has an unusually large file size compared with the original version, it may require further investigation.

However, **unusual file size alone does not prove that hidden data exists**.

Step 3 – Examine Metadata

Image metadata can contain information such as:

- Software used to create the image
- Date and time
- Camera information
- Author information
- Comments
- GPS information, depending on the file

Investigators can compare the suspicious logo's metadata with a known legitimate copy.

Unexpected metadata may provide an investigative clue.

Important Point

Metadata anomalies are **indicators, not proof**, because legitimate software can also modify or add metadata.

Step 4 – Analyze the Image for Steganography

A common form of image steganography hides information in the **least significant bits (LSBs)** of pixels.

For example, a pixel value might be represented in binary as:

```
10110110
```

If the last bit is changed:

```
10110111
```

the numerical difference is very small and may not be noticeable visually.

If this process is performed across many pixels, hidden information may be stored while keeping the image visually similar.

Step 5 – Perform Pixel and Statistical Analysis

Investigators can examine the image's pixel patterns.

Possible methods include:

1. LSB Analysis

The investigator examines the least significant bits of pixel values to determine whether they contain unusual patterns.

2. Histogram Analysis

A histogram shows the distribution of pixel values.

A suspicious image can be compared with:

- A known original image
- An image generated using the same software
- Expected statistical characteristics

Unexpected statistical patterns may indicate possible manipulation.

3. Color Channel Analysis

RGB images contain three channels:

- Red
- Green
- Blue

Investigators can examine each channel separately to identify unusual patterns.

Step 6 – Compare With the Original Image

If an authentic company logo is available, investigators should compare it with the suspicious image.

For example:

```
Known Original Logo
        +
  Suspicious Logo
        ↓
Pixel-by-pixel comparison
        ↓
Identify unexpected differences
```

If the images appear identical to the human eye but have systematic pixel-level differences, this may indicate hidden modifications.

Step 7 – Check the File Structure

Investigators can also inspect the internal structure of the file.

For example, they may look for:

- Unexpected data after the normal end of the image
- Unusual file sections
- Embedded archives

- Unexpected strings
- Inconsistent file headers
- Data that does not correspond to the declared image format

A file that claims to be an image but contains suspicious additional structures deserves further examination.

Step 8 – Use Steganalysis Tools

Specialized forensic and steganalysis tools can assist investigators.

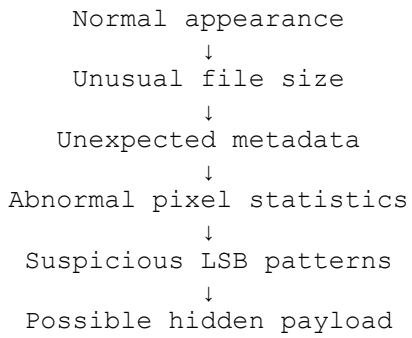
Examples include:

- **ExifTool** – examines metadata.
- **binwalk** – helps analyze file structures and embedded data.
- **strings** – identifies readable strings within a file.
- Dedicated steganalysis tools – can perform statistical and LSB analysis.

These tools should be used as part of a broader investigation rather than treating one tool's output as conclusive evidence.

Possible Investigation Findings

Suppose the investigators find:



This combination provides stronger evidence that the image may have been modified for hidden communication.

The investigators would then document the evidence and, where authorized, attempt to identify or extract the concealed data.

Two Techniques to Strengthen Hidden Communication Against Detection or Extraction

If the purpose is to understand how such communication can be made more resistant to forensic detection, two relevant techniques are:

Technique 1 – Encrypt the Hidden Data Before Embedding

Instead of directly embedding readable confidential information, the sender can first encrypt the information.

The process becomes:

```
Confidential Message
        ↓
    Encryption
        ↓
    Ciphertext
        ↓
Steganographic Embedding
        ↓
      Image
```

For example:

```
"CONFIDENTIAL"
        ↓
    Encryption
        ↓
  Encrypted Data
        ↓
Embedded into Image
```

Advantage

Even if investigators successfully extract the hidden data, they may obtain ciphertext rather than the original plaintext.

Therefore, this provides **two layers of protection**:

1. Steganography hides the existence of the message.
2. Encryption protects the content of the message.

Important Limitation

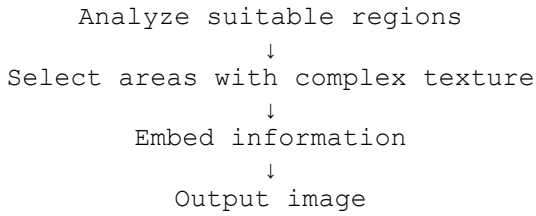
Encryption does **not** prevent investigators from detecting that an image has been modified. It primarily protects the **content** after extraction.

Technique 2 – Use Adaptive Steganographic Embedding

Instead of embedding hidden information uniformly throughout the image, adaptive steganography selects image regions where modifications are less statistically noticeable.

For example:

```
Image
  ↓
```



Complex regions can naturally contain more variation than smooth regions, so small changes may be harder to distinguish visually and statistically.

Advantage

It can reduce some detectable statistical artifacts compared with simple, predictable embedding methods.

Limitation

No steganographic technique is guaranteed to remain undetected. Advanced steganalysis can still identify suspicious statistical patterns.

Comparison of the Two Protection Techniques

Technique	Main Purpose	Benefit	Limitation
Encrypt data before embedding	Protect message content	Extracted data remains encrypted	Does not necessarily hide the existence of steganography
Adaptive embedding	Reduce detectable embedding artifacts	Can make statistical detection more difficult	Advanced steganalysis may still detect manipulation

Security Analysis

The security of hidden communication can be considered at two different levels:

Level 1 – Detectability

Can investigators determine that an image has been modified?

This is addressed by **steganographic techniques and careful embedding**.

Level 2 – Extractability/Confidentiality

If investigators discover and extract the hidden information, can they understand it?

This is addressed by **encrypting the hidden payload before embedding**.

Therefore:



```
Hide existence          Protect content
      ↓                       ↓
 Steganography           Encryption
```

Using both provides defense at different stages of the investigation.

Result

The suspicious company logo can be investigated using a combination of **file-property examination, metadata analysis, pixel/LSB analysis, histogram analysis, file-structure examination, comparison with an authentic copy, and specialized steganalysis tools**.

A visually normal image does not necessarily mean that it contains no hidden information.

Two techniques that can strengthen hidden communication are:

1. **Encrypting the confidential data before embedding it.**
2. **Using adaptive steganographic embedding to reduce obvious statistical artifacts.**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation